GEHU/02E/9.1.3

Graphic Era HILL UNIVERSITY

Established by an Act of the State Legislature of Uttarakhand (Adhiniyam Sankhya 12 of 2011) University under section 2(f) of UGC Act, 1956

End Term (Even) Semester Regular/Back Examination May 2026

Roll no.......................

Name of the Program : BCA/BCA AI/DS / B. Sc. IT
Semester: 04
Name of the Course : Web Application Development
Course Code: TBC /TBD 401 / TBI 401
Time: 3 hours

Maximum Marks: 100

Note:

(i) All the questions are compulsory.
(ii) Answer any two sub questions from a, b and c in each main question.
(iii) Total marks for each question is 20 (twenty).
(iv) Each sub-question carries 10 marks.

Q1. (2X10=20 Marks)

a. Explain the concept of hyperlinks in XHTML. Write an XHTML program that demonstrates the use of the anchor (<a>) tag, including attributes such as href, target, and title, with suitable examples. CO1

b. Explain the role of web browsers in accessing internet resources. Discuss the different types of web browsers and describe how a browser communicates with a web server to retrieve and display a web page. CO1

c. Write an XHTML program to demonstrate different types of lists. Include ordered list, unordered list, and definition list with suitable examples. CO1

Q2. (2X10=20 Marks)

a. Explain the different types of CSS, Inline CSS, Internal CSS, and External CSS. Write suitable examples to demonstrate each type. CO2

b. Design an XHTML form for user feedback or login/registration. Also explain the purpose of each form element used. CO2

c. Explain the concept of the CSS Box Model. Describe its components, namely content, padding, border, and margin, with a suitable diagram and example. CO2

Q3. (2X10=20 Marks)

a. Design an XHTML form and write JavaScript code to perform form validation. Validate the following:

- Name (should not be empty)
- Email (valid format)
- Password (minimum 6 characters)
- Mobile number (only digits, 10 characters)

Display appropriate error messages and prevent form submission if validation fails. CO3

b. Explain the concept of events and events registration in JavaScript. Write a program to demonstrate different types of events such as:

- onclick
- onmouseover
- onchange

CO3

c. Write a JavaScript program to create an array named course using different methods and initialize it with the elements "BCA", "MCA", "BSCIT", and "MSCIT". Perform the following operations: CO4

1. Access and display array elements
2. Modify an existing element (e.g., change "MCA" to "MBA")

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3. Insert a new element (e.g., "B.Tech")
4. Delete an element from the array
5. Traverse the array using loops (for, forEach)

Q4. (2X10=20 Marks)

a. Explain different types of arrays in PHP (indexed, associative, and multidimensional). Write a program demonstrating operations on each type. CO5

b. Write a PHP program to perform CRUD operations (Create, Read, Update, Delete) using a MySQL database. Create a sample table (e.g., student with fields like id, name, course) and demonstrate all four operations with proper database connectivity. CO5

c. xplain the concept of string handling in PHP. Write a PHP program demonstrating at least four string functions such as strlen(), strrev(), strtoupper(), strpos(), etc. CO4

Q5. (2X10=20 Marks)

a. Discuss the advantages of using the MERN Stack for web development. Explain why it is widely adopted in modern web applications, with relevant examples. CO6

b. Explain how different components of the MERN stack work together in a typical application. Describe the flow of data from the user interface to the database and back. CO6

c. Define Full Stack Web Development. Explain the roles of the front-end, back-end, and database layers in building a web application, with suitable real-world examples. CO5

Note For the question paper setters:

- ~~Question paper should cover all the COs of the course, preferably equal distribution.~~
- ~~Please specify COs against each question.~~